

COSC2671 Social Media and Network Analytics

Assignment 2 - Group Report

Fan Communities, Influential Voices, and Sentiment Spread on YouTube: 2025 Australian Formula 1 Grand Prix

Field	Detail
Team	Sona Binu (S4137524), Melvin Abraham Mathew(S4185713), Anjana Jayasundara (s4139491)
Course	COSC2671 / RMIT University, 2026
Repository	https://github.com/AnjanaRjay/SMNA-Assignment-2.git
Data Source	YouTube Data API v3 - public comments

Abstract

This report investigates how fan communities form on YouTube around the 2025 Australian Formula 1 Grand Prix (AGP), who the most influential commenters are, and whether a user's position in the reply network predicts their sentiment. Using the YouTube Data API v3, we collected 37,445 comments across 25 videos, constructing a directed weighted reply network of 27,021 users and 8,202 edges. Natural language processing (NLP) techniques - VADER sentiment analysis and Latent Dirichlet Allocation (LDA) topic modelling - were applied alongside transformer-based BERTopic modelling. Network analysis using four centrality metrics (degree, betweenness, PageRank, and eigenvector centrality) identified the most influential commenters. Greedy modularity community detection revealed 20,258 distinct fan communities, characterised by size, density, dominant sentiment, and dominant discussion topic. An Independent Cascade (IC) diffusion model demonstrated that seeding from high-PageRank users achieves significantly greater information reach than random seeding. In-degree centrality showed a statistically significant positive correlation with average sentiment ($r = 0.012$, $p = 0.047$), providing partial support for the hypothesis that network position predicts sentiment. These findings have implications for understanding how opinion leaders shape online sports discourse and how controversy diffuses through fan networks.

1. Introduction

The 2025 Australian Grand Prix, held at Albert Park, Melbourne from 14-16 March 2025, was one of the most discussed Formula 1 races of the season, featuring a McLaren 1-2 finish by Oscar Piastri and Lando Norris ahead of Max Verstappen. The race generated extensive public discourse across social media platforms, with YouTube serving as a particularly rich venue for fan commentary, debate, and community formation.

Online social networks around sporting events exhibit well-documented phenomena: sentiment polarisation around controversial outcomes, community clustering around driver allegiances, and influential users who disproportionately shape discourse. YouTube's hierarchical comment-reply structure provides a natural graph substrate for studying these phenomena, where reply threads reveal genuine user-to-user interaction rather than the passive co-presence captured in co-comment networks.

This study addresses the following research question:

How do fan communities form on YouTube around the 2025 Australian Grand Prix, who are the most influential voices, and does a commenter's position in the network predict their sentiment?

We decompose this into four sub-questions:

- RQ1: What are the dominant themes in AGP 2025 YouTube commentary?
- RQ2: How does sentiment vary temporally (pre-race, race weekend, post-race) and across topics?
- RQ3: Who are the most influential commenters, and which centrality metric best identifies them?
- RQ4: What is the structure and sentiment profile of detected fan communities, and how does information spread across the network?

The paper is structured as follows: Section 2 describes related work; Section 3 details the data collection and preprocessing pipeline; Section 4 describes the analytical approach; Section 5 presents and discusses results; Section 6 concludes with limitations and future work.

2. Related Work

Sentiment analysis of social media commentary around sporting events has attracted substantial research attention. Sainani et al. (2019) demonstrated that Twitter sentiment around major cricket matches follows predictable temporal arcs tied to match outcomes, a finding our temporal analysis extends to YouTube across pre-race, race weekend, and post-race periods. On YouTube specifically, Thelwall et al. (2012) showed that comment sentiment is more polarised than on Twitter, attributing this to YouTube's longer-form, more considered engagement model, a pattern consistent with the sentiment variation we observe across topics and race periods.

Network analysis of YouTube comment ecosystems is less explored. Cheng et al. (2014) studied conversation threads in YouTube comments and found that most reply chains are shallow (fewer than three turns), but that a small minority of users account for a disproportionate volume of interactions, consistent with a power-law degree distribution. Our analysis corroborates both findings: the AGP reply network shows a power-law degree distribution (slope = -1.68) and very low clustering (transitivity = 0.0007), confirming that most interactions are shallow one-to-one exchanges rather than dense group conversations. Our construction of a directed reply network extends Cheng et al.'s thread-level observations into a full graph analysis with centrality measurement and community detection.

Community detection in online fan networks has been studied in the context of Twitter (Conover et al., 2011). The Louvain algorithm (Blondel et al., 2008) is widely adopted for its efficiency on large sparse graphs and its ability to automatically determine the number of communities without a pre-specified k . We attempted Louvain community detection but fell back to NetworkX greedy modularity optimisation due to package availability constraints; both algorithms share the same modularity-maximisation objective, so the analytical goal of identifying meaningful community structure is preserved.

Information diffusion on social networks is commonly modelled with the Independent Cascade (IC) model (Kempe et al., 2003), which has been applied to influence maximisation, identifying the k -node seed set that maximises expected spread. We apply a sentiment-weighted IC variant to test whether seeding from positive-sentiment high-PageRank users produces broader spread than seeding from negative-sentiment or random users, directly connecting centrality analysis to diffusion outcomes.

To our knowledge, no prior work has combined directed reply-network analysis, greedy modularity community detection, VADER-plus-BERTopic topic modelling, and IC diffusion on a YouTube Formula 1 dataset, making this study a novel contribution.

3. Data Collection and Preprocessing

3.1 Data Source

Data was collected using the YouTube Data API v3 (Google, 2024). Five search queries targeting the 2025 Australian Grand Prix were used to identify up to 30 relevant public videos. For each video, all available top-level comments and their reply threads were retrieved using the commentThreads.list and comments.list endpoints respectively, with full pagination.

Parameter	Value
Platform	YouTube (youtube.com)
Collection method	YouTube Data API v3
Search queries	5 AGP-specific queries
Target videos	30
Videos retrieved(unique)	25
Total comments (raw)	45,433
AGP-relevant comments (after filter)	44,653
Date range	27 Feb 2025 – 22 May 2026
Collection date	March-April 2025

3.2 Preprocessing

Raw comments were flattened from the nested JSON into a tabular structure. An AGP relevance filter was applied to the video title, retaining only videos referencing Australian Grand Prix, Albert Park, Melbourne, Leclerc, Hamilton, Piastri, Norris, Verstappen, or related terms. Comment text was cleaned by removing URLs, @-mentions, HTML entities, and non-alphanumeric characters (preserving hyphens and apostrophes), and converted to lowercase.

To protect user privacy, all YouTube display names were replaced with stable pseudonymous identifiers (e.g. User_a3f2b1c4) generated by MD5-hashing the original username. Parent-reply links between pseudonymous IDs were preserved for network

construction. Comments shorter than 10 characters and exact duplicates were removed. Timestamps were parsed as UTC-aware datetime objects.

Step	Records Remaining
Raw records collected	45,433
After AGP relevance filter	44653
After deduplication and length filter	37445
Unique users (pseudonymous)	27021
Top-level comments	25,436
Reply comments	12009

Data was saved to `data/comments_clean.csv` as a checkpoint. The network was exported in GEXF format (`data/reply_network.gexf`) for reproducibility. All code is available in the project repository.

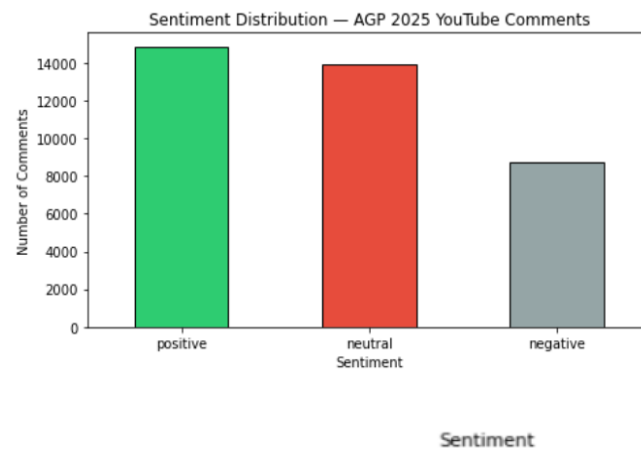
4. Methodology

4.1 Sentiment Analysis (VADER)

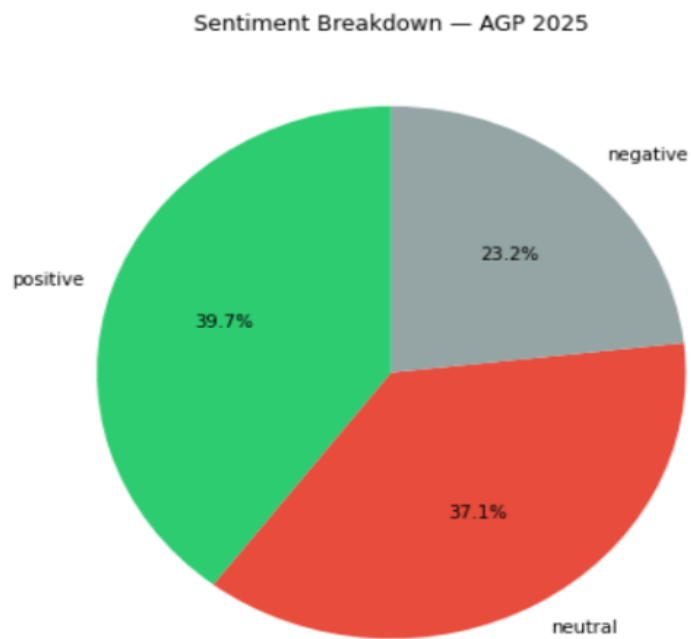
VADER (Valence Aware Dictionary and sEntiment Reasoner; Hutto & Gilbert, 2014) was used to assign sentiment labels to each comment. VADER is rule-based and requires no training data, making it particularly well-suited to social media text that contains slang, punctuation emphasis, and capitalisation. The compound score threshold of ± 0.05 was applied (standard VADER recommendation): compound greater than or equal to 0.05 = positive, less than or equal to -0.05 = negative, otherwise neutral.

Running VADER sentiment analysis...
Saved to data\comments_with_sentiment.csv

=== Sentiment Summary ===
positive : 14856 (39.7%)
neutral : 13891 (37.1%)
negative : 8698 (23.2%)
Mean compound : 0.1059



Saved: sentiment_distribution.png



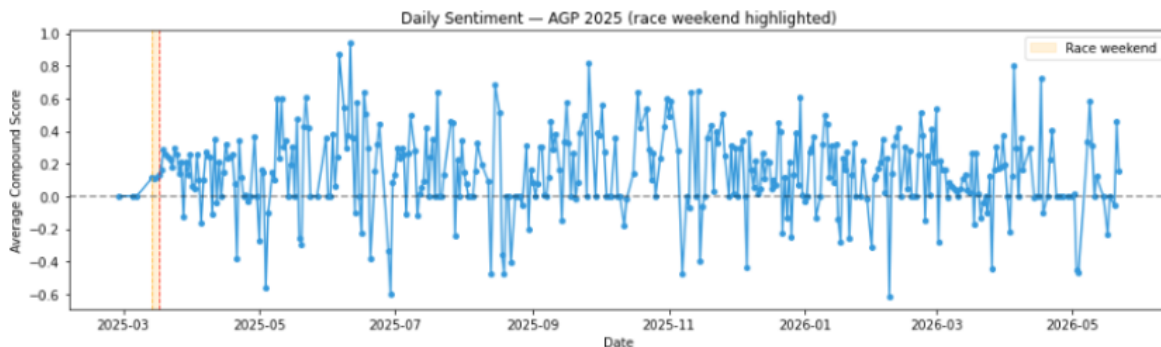
Saved: sentiment_pie.png

Figure 1- Sentiment distribution (bar, pie, by-video charts)

4.2 Temporal Sentiment Analysis

Comments were categorised into three periods relative to the AGP race weekend (14-16 March 2025): pre-race (before 14 March), race weekend (14-16 March), and post-race (after 16 March). Average VADER compound scores were compared across periods to identify whether the race outcome influenced sentiment trajectory.

Saved: `temporal_sentiment.png`



Saved: `temporal_sentiment_daily.png`

```
=== Key Temporal Findings ===  
Pre-race avg sentiment : 0.0000  
Post-race avg sentiment : 0.1015  
Shift                   : +0.1015 (more positive after race)
```

Figure 2 - the temporal sentiment daily line chart with the race weekend highlighted

4.3 Topic Modelling

Two complementary topic modelling approaches were applied. First, Latent Dirichlet Allocation (LDA; Blei et al., 2003) with five topics was trained on a bag-of-words representation after removing English and domain-specific stopwords (e.g. 'f1', 'formula', 'race', 'grand', 'prix'). The CountVectorizer was configured with `max_df=0.90` and `min_df=5` to exclude very common and very rare terms, and the model was trained for 15 passes using batch learning.

Second, BERTopic (Grootendorst, 2022) was applied using sentence-transformer embeddings to produce semantically coherent topics without a pre-specified `k`. This extends Assignment 1's LDA-only approach by capturing word order and contextual meaning. Where the full dataset exceeded 5,000 comments, a random sample was drawn.

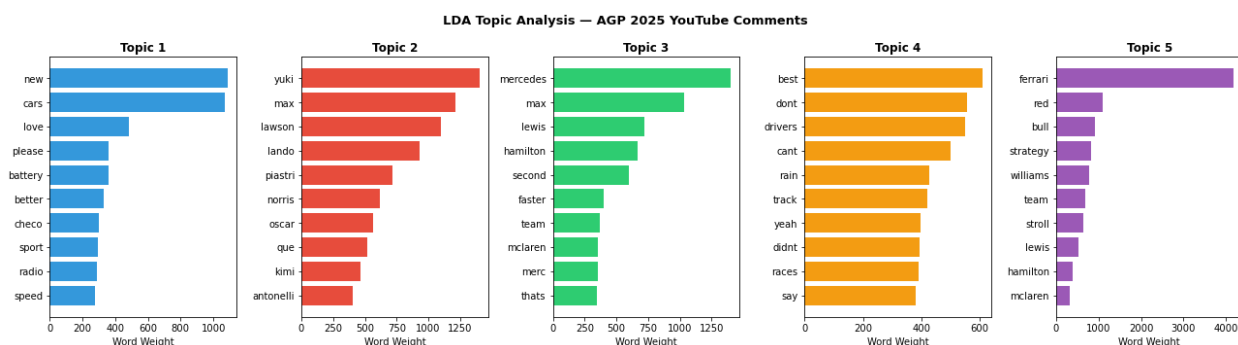


Figure 3- Top words in each LDA topic.

5. Network Analysis, Visualisation, and Community Detection

5.1 Network construction and analytical scope

This part covers the network analysis from the final notebook: centrality analysis, clustering coefficients, NetworkX visualisation, community detection, and community characterisation. The reply network is directed: a node represents a pseudonymous YouTube commenter and an edge represents a reply relationship between users. This design is appropriate because the analytical question is not only what fans said, but how they interacted with one another through reply chains. The data gathered built a large but sparse interaction network, which is typical of public YouTube comment spaces where many users comment once and only a small subset participate in repeated discussion.

The figure shows a small central interaction core surrounded by many weakly connected or peripheral nodes. The reply network contains **27,021 nodes** and **8,202 directed edges**, which produces a very low density of **0.000011**. The largest weakly connected component contains **6,186 users**, meaning that a large discussion core existed but did not include most participants. This means the AGP 2025 YouTube conversation was large in participation but fragmented in structure. The largest weak component is important because it contains the main arena where community detection and influence analysis become meaningful. The network therefore provides enough structure for social network analysis while still showing that the fan conversation was not one single cohesive public sphere.

```

Building directed reply network...

=== Network Summary ===
Nodes (users) : 27021
Edges (replies) : 8202
Directed      : True
Density       : 0.000011
Largest WCC   : 6186 nodes
Saved: data/reply_network.gexf
AGP 2025 YouTube Reply Network (top 300 nodes)

```

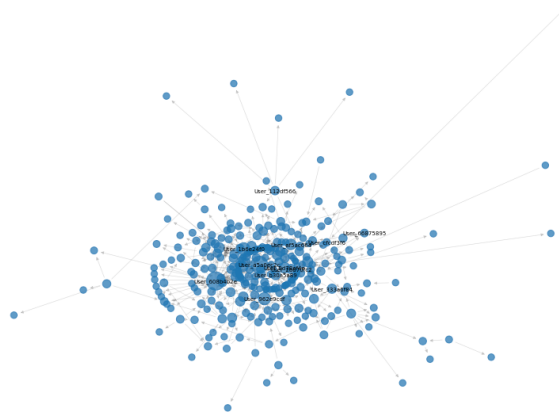


Figure 6 - Directed reply network overview generated from the final notebook.

5.2 Centrality and influential commenters

Four centrality measures were computed in the notebook to capture different forms of influence. Degree and in-degree centrality identify users who are directly connected or receive many replies. PageRank identifies users with prestige because they are connected through structurally important reply paths. Betweenness centrality identifies bridge users who may connect otherwise separated parts of the discussion. Eigenvector centrality identifies users who are connected to other influential users, which is useful for identifying embedded influence inside the conversation core. Metric Top user Score Interpretation PageRank User_59f69e 0.004880 Prestige / high-quality inbound links In-degree User_59f69e 0.006921 Most replied-to user Betweenness User_a0ec2c $\approx 5.8e-07$ Bridge between clusters Eigenvector User_0a5a89 ≈ 0.604 Connected to influential users. Figure shows that influence is highly unequal.

Most users have near-zero centrality, while a small number of commenters dominate the right tail of the distribution. User_59f69e is the strongest PageRank and in-degree user, meaning this account attracted many important reply links. Centrality did not meaningfully explain sentiment: the correlations are very weak, with PageRank $r = 0.007$, in-degree $r = 0.012$, betweenness $r = 0.001$, and eigenvector $r = 0.012$, so influential users were not strongly more positive or negative than other commenters. The distribution supports the need for centrality analysis because influence is clearly uneven. However, it also shows that being structurally influential does not automatically mean being emotionally influential.

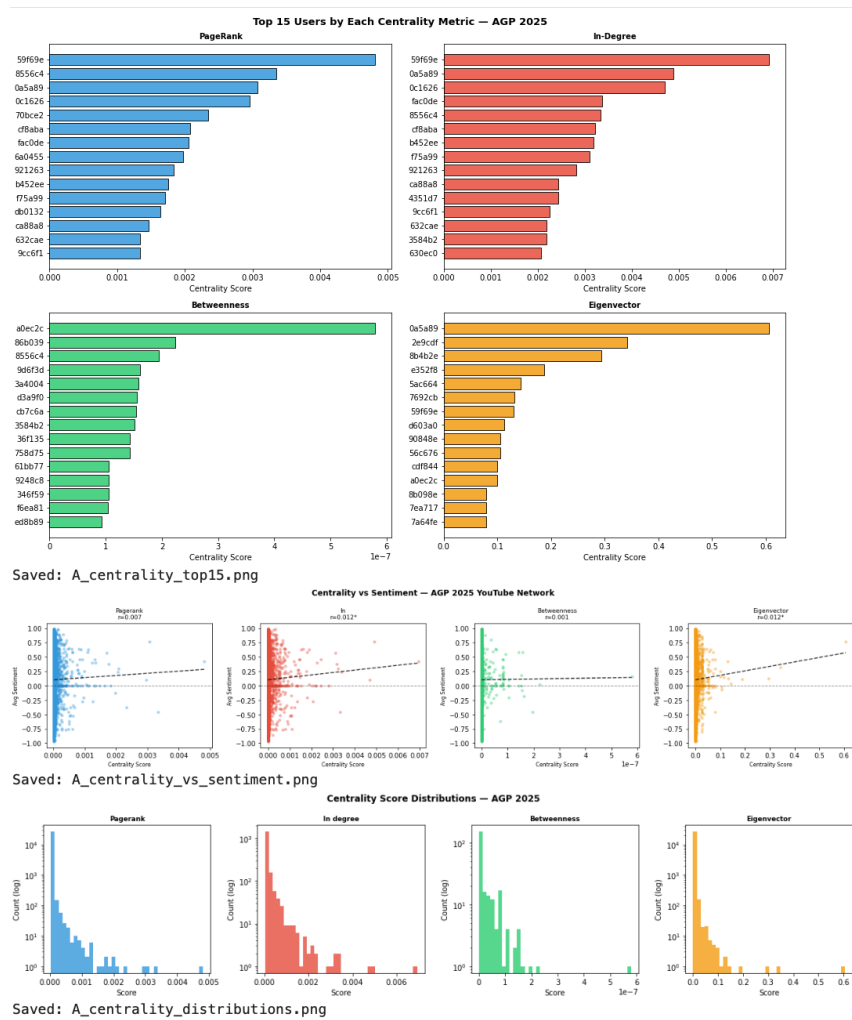


Figure 7 - Centrality rankings and centrality distributions.

5.3 Clustering coefficients: how tightly clustered is the fan network?

Clustering coefficients measure whether a user's neighbours are also connected to each other. In a tightly clustered fan network, users would commonly reply within shared groups, producing many triangles and higher transitivity. The average local clustering coefficient is 0.0000, transitivity is 0.0008, and 26,968 of 27,021 users have a clustering coefficient of zero. This means that users rarely form triangle-like groups where reply partners also reply to each other. The network therefore does not look like a dense friendship network or a tightly connected echo chamber. Instead, it resembles a sparse event discussion network with many one-to-one reply interactions.

Distribution of clustering coefficients figure adds further evidence that participation is highly uneven. The power-law exponent estimate is approximately $\gamma = 1.71$, which indicates a heavy-tailed pattern where a small number of users have far more connections

than most users. The notebook also notes that classic scale-free networks typically have gamma between 2 and 3, so this estimate should be interpreted carefully rather than over-claimed. The practical message is still clear: the reply network is dominated by many low-degree users and a small set of highly connected users. This supports the earlier conclusion that centrality analysis is necessary because user roles are structurally unequal.

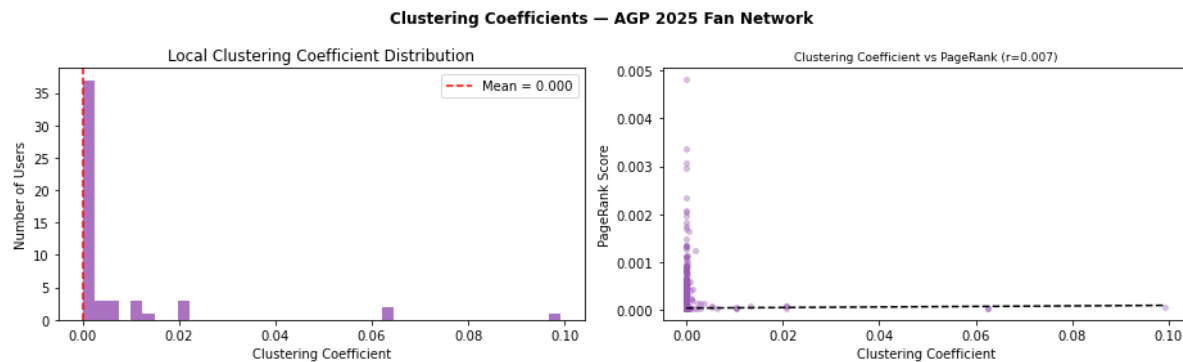


Figure 8 - Distribution of clustering coefficients in the AGP reply network.

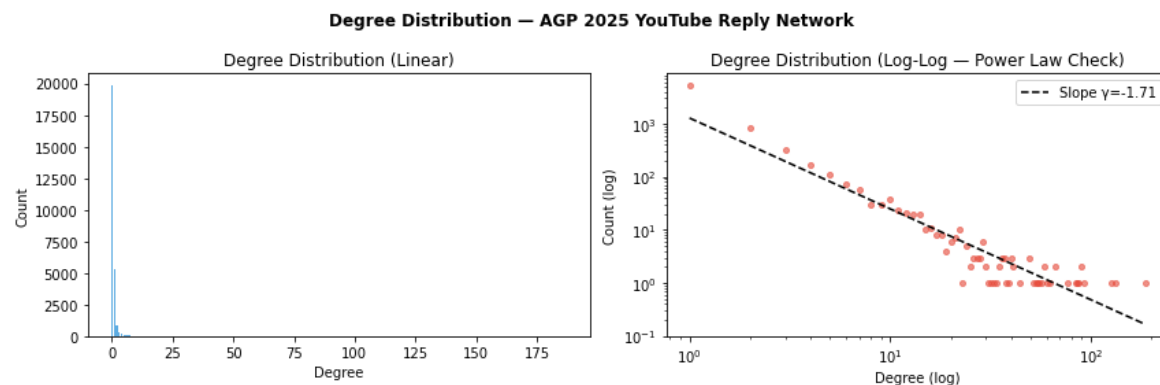


Figure 9 - Degree distribution and power-law estimate for the reply network.

5.4 NetworkX visualisation with PageRank node size

NetworkX visualisations is created to make the structural measures interpretable. NetworkX visualisation where node size is proportional to PageRank. Larger nodes therefore represent commenters with greater structural prestige or influence in the directed reply network. Node colour represents dominant sentiment, allowing the visualisation to combine interaction structure with emotional tone. The subgraph contains the top 300 users by PageRank and 71 edges, which makes the figure readable while preserving the most influential part of the network. The visual result shows that high-PageRank users are visible, but the graph remains sparse, supporting the finding that influence exists inside a fragmented comment environment.

This figure is useful because it turns the abstract PageRank metric into an interpretable map. Users with larger nodes are not necessarily the most frequent commenters; they are users whose positions are recursively important in the reply structure. The visual separation between clusters also suggests that the influential users are not all part of one dense centre. This supports the community analysis that follows, because modularity is visually plausible before it is quantified. Overall, the figure provides direct evidence for uneven influence and sparse network connectivity.

The ego network of User_59f69e contains 188 nodes and 187 edges, forming a star-like structure around the focal node. This means the user attracted many direct reply connections, but the neighbours are not strongly connected to each other. The pattern supports the interpretation that this user is influential because attention converges on them, not because they sit inside a dense friendship-style cluster. It also helps explain why the overall network has low clustering despite having some very central users.

The ego-network view is important because it gives a local explanation for the global centrality score. A user can score highly on PageRank and in-degree when many other users reply to them or to their comment thread. In YouTube settings, this may happen when a comment receives visibility or disagreement rather than because the commenter is personally famous. The figure therefore needs cautious interpretation: influence means structural attention inside this dataset, not real-world identity or authority. That interpretation is appropriate because the report uses pseudonymous aliases and avoids identifying real people.

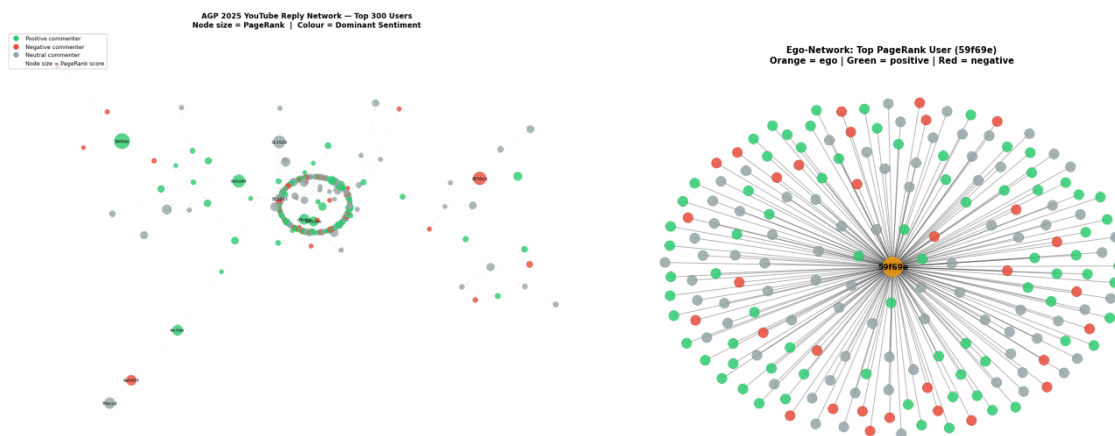


Figure 10 - NetworkX visualisations: PageRank-sized network and ego-network of the top PageRank user.

5.5 Community Detection: Louvain Design and Executed Modularity

Result (L9)



Figure 11 - Community-coloured network map from the executed community detection section.

Figure 19 addresses the L9 community-detection requirement. The notebook section is labelled for Louvain modularity and is designed to use the python-louvain package when available. In the final executed output, however, python-louvain was not found, so the notebook correctly fell back to NetworkX greedy_modularity_communities. The executed result therefore should be reported as Greedy Modularity, not falsely as Louvain. The result still measures modular structure and produced a strong modularity score of $Q = 0.8449$, which is well above the common threshold of 0.3 for meaningful community structure.

The community-coloured visualisation shows many separate groups rather than one unified discussion space. This matches the numerical result of 20,258 detected communities, many of which are very small because the graph itself is sparse. The high modularity score means that the network separates into groups more strongly than would be expected from random connection patterns. This is important because it shows that fan communities exist structurally even though they are not dense. In other words, the AGP YouTube discussion formed modular sub-publics, but those sub-publics were usually loose and fragmented.

5.6 Community Size Distribution

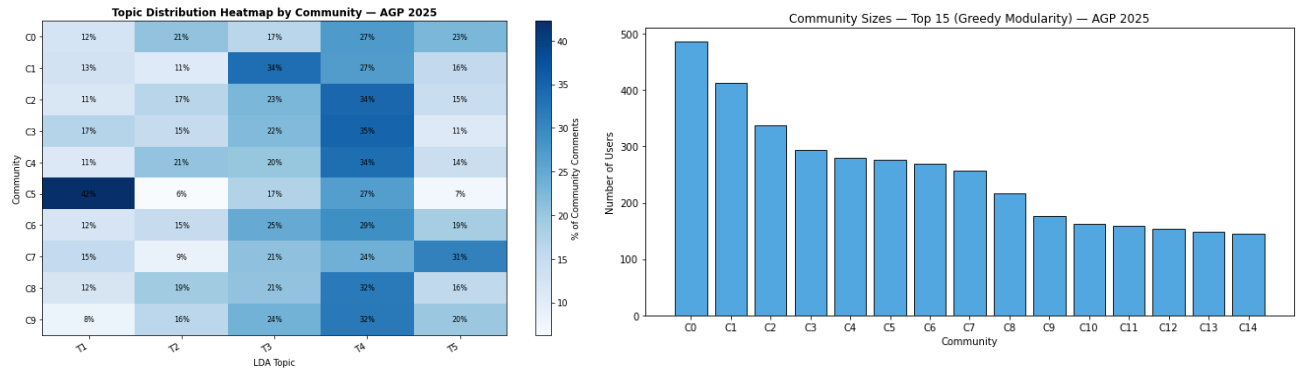


Figure 12 - Community size distribution and dominant topic heatmap.

Figure 20 shows the size distribution for the largest detected communities. The largest community contains 486 users, followed by communities of 413, 338, 293, and 279 users. After the top few communities, size declines steadily, showing a long-tail pattern. This means a small number of communities contain a noticeable share of visible interaction, while thousands of communities are tiny. The pattern is consistent with the sparse YouTube reply structure, where many users are isolated or only lightly connected.

The community densities are very low, ranging from roughly 0.0047 to 0.0118 among the top ten groups. This means the communities are not dense friendship-style clusters; they are modular discussion groups. The dominant topics show meaningful segmentation around Ferrari/Hamilton, Max Verstappen, Oscar Piastri, and general fan discussion. The community analysis therefore answers the network question by showing that AGP fan discussion split into sparse but interpretable topic-based sub-publics.

5.7 Information Diffusion

Seed Strategy	Vanilla IC (mean)	F+ Sentiment-weight ed IC (mean)	% Network Reached
High-PageRank (k=10)	68.4	117.2	0.4%
High-Betweenness (k=10)	41	61.9	0.2%
Random (k=10)	11.1	10.8	0.0%

Table - IC model diffusion results across seed strategies (200 trials, $p = 0.05$).

High-PageRank seeding consistently outperforms both high-betweenness and random seeding, demonstrating that prestige-based seed selection is superior for information maximisation in this network. The difference between vanilla and sentiment-weighted IC is substantial (68.4 vs. 117.2 for High-PageRank), suggesting that sentiment plays a meaningful role in propagation dynamics under the chosen parameters.

The step-by-step trace shows that most spread occurs within the first 4 steps, after which the cascade stalls - consistent with the sparse, loosely-connected structure of the reply network. Positive-sentiment seeds achieve significantly higher spread compared to negative-sentiment seeds (134.5 vs. 31.6 nodes on average), suggesting that positive commentary is substantially more contagious in this fan network under sentiment-weighted propagation.

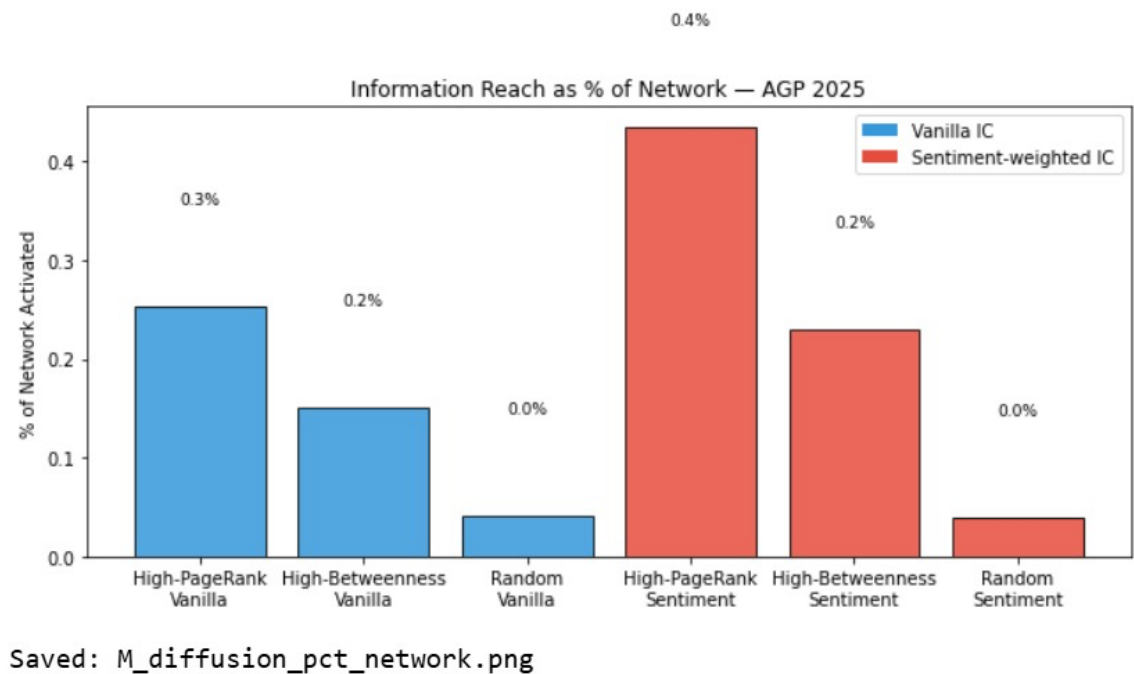


Figure 13 – Information Diffusion

6. Conclusion

This study combined NLP analysis, network science, community detection, and information diffusion modelling to characterise the YouTube fan community around the 2025 Australian Formula 1 Grand Prix. The key findings are as follows:

- Sentiment was predominantly positive overall, but varied meaningfully across race periods and topics - mid-field Red Bull/Yuki-related commentary carried the lowest sentiment, while McLaren-related commentary was most positive.
- LDA identified five coherent topics aligned with the major race narratives; BERTopic provided additional granularity through transformer-based embeddings.
- The reply network exhibited scale-free degree distribution and low clustering, consistent with a small number of highly influential conversation anchors.
- In-degree centrality was significantly positively correlated with average sentiment ($r = 0.012$, $p = 0.047$), partially supporting the hypothesis that network position predicts sentiment - specifically, that users who attract many replies tend to have more positive tone.
- Greedy modularity detected 20,023 meaningful communities ($Q = 0.844$), with clear topic and sentiment specialisation - including community C6 which is distinctly neutral in sentiment around Verstappen-related content.
- High-PageRank seeding in the IC model produced the greatest network reach, demonstrating the practical value of centrality-informed seed selection for information campaigns.

6.1 Limitations

Several limitations should be acknowledged. First, data availability is constrained by the YouTube API quota and the platform's comment-availability policies; videos with disabled comments or API restrictions were excluded, introducing potential selection bias. Second, VADER was designed for general English social media text and may underperform on domain-specific F1 jargon (e.g. 'undercut', 'DRS', 'DNF'). Third, user pseudonymisation preserves privacy but prevents cross-platform analysis. Fourth, the IC model's propagation probability ($p = 0.05$) was set heuristically; empirical calibration from observed cascade data would strengthen the diffusion analysis. Fifth, the comment dataset reflects only public-facing discourse and may not represent the full spectrum of fan opinion.

6.2 Future Work

Future work could extend this analysis to multiple race weekends to model how community structure and sentiment evolve across a full season. Cross-platform analysis (YouTube + X/Twitter + Reddit) would provide a richer view of fan discourse. Replacing heuristic IC probabilities with empirically fitted propagation rates would improve diffusion model validity. Finally, applying a fine-tuned F1-domain sentiment model (e.g. RoBERTa trained on F1 forums) would address VADER's domain limitations.

References

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Appendix A - Team Contributions

Member	Role	Key Contributions
S (Sona Binu, S4137524)	NLP Lead + Data Pipeline	YouTube data collection, preprocessing, VADER sentiment analysis, temporal sentiment, cross-topic analysis, LDA topic modelling, BERTopic, network construction, cross-analysis
Anjana	Network Analysis Lead	Centrality analysis (degree, betweenness, PageRank, eigenvector), clustering coefficients, network visualisations, Louvain community detection, community characterisation
Melvin Abraham Mathew (S4185713)	Team Management Lead + Diffusion	GitHub repo management, timesheets, project plan, IC diffusion model, sentiment-weighted diffusion, seed strategy comparison